# Lab Session 03

## *Examine effects of quantization on discrete time continuous signals*

## Quantization:

Digital signal is a sequence of numbers (samples) in which each number denotes a finite number having finite precision.

The process of converting a discrete time continuous valued (DTCV) signal into a discrete time discrete valued (DTDV) signal is known as Quantization. In this process each sample expresses a value as finite (rather than infinite) number. The error introduced in representing the continuous valued signal by a finite set of discrete value levels is called Quantization error or Quantization noise.

We express the quantizer operation on the sampled signal x(n) as Q[x(n)] and the quantized signal can be expressed as xq(n) which denotes the sequence of quantized samples at the quantizer output. Hence

$$x_q = Q[x(n)] \qquad (3.1)$$

The quantization error is a sequence eq(n) defined as the difference between the quantized value and the actual sample value. Thus

$$e_q(n) = x_q(n) - x(n) \qquad (3.2)$$

It is obvious that the signal cannot be processed by using calculator or digital computer since only the first few samples can be stored and manipulated. For example, most calculators process number with only eight significant digits.

We assume that we want to use only one significant digit. To eliminate the excess digit we either simply discard them (truncation) or discard them by rounding the resulting number (rounding). The resulting quantized signals $x_q(n)$ are shown by the numerical illustrations in the Table 3.1 as below:

| n | x(n) Discrete Time Signal | xq(n) Truncation | xq(n) Rounding | eq(n) = xq(n)-x(n) |
|---|---|---|---|---|
| 0 | 1 | 1.0 | 1.0 | 0.0 |
| 1 | 0.9 | 0.9 | 0.9 | 0.0 |
| 2 | 0.81 | 0.8 | 0.8 | -0.01 |
| 3 | 0.729 | 0.7 | 0.7 | -0.029 |
| 4 | 0.6561 | 0.6 | 0.7 | 0.0439 |
| 5 | 0.59049 | 0.5 | 0.6 | 0.00951 |
| 6 | 0.531441 | 0.5 | 0.5 | -0.031441 |
| 7 | 0.4782969 | 0.4 | 0.5 | 0.0217031 |
| 8 | 0.43046721 | 0.4 | 0.4 | -0.03046721 |
| 9 | 0.387420489 | 0.3 | 0.4 | 0.012579511 |

Table 3.1: Error Computation and Analysis

The values allowed in the digital signal are called the quantization levels whereas the

distance Δ between two successive quantization levels is called quantization step size or resolution. The rounding quantizer assigns each sample of x (n) to the nearest quantization level. In contrast, a quantizer that performs truncation would have assigned each sample of x(n) to the quantization level below it.

The quantization error eq (n) in rounding is limited to the range of -Δ/2 to Δ/2, that is;

$$-\frac{\Delta}{2} \leq e_q(n) \leq \frac{\Delta}{2} \qquad (3.3)$$

The instantaneous quantization error cannot exceed half of the quantization step.

If $x_{min}$ and $x_{max}$ represent the minimum and maximum values of x (n) and L is the number of quantization levels, then

$$\Delta = \frac{X_{max} - X_{min}}{L-1} \qquad (3.4)$$

The dynamic range of the signal is $X_{max}$-$X_{min}$. In our example we have $X_{max}$ = 1, $X_{min}$ = 0, and L = 11, which leads to Δ = 0.1. Note that if the dynamic range is fixed, increasing the number of quantization levels L results in a decrease of the quantization step size, thus the quantization error decreases and the accuracy of the quantizer increases. In practice, we can reduce the quantization error to an insignificant amount by choosing a sufficient number of quantization levels.

## SQNR (signal to quantization ratio)

The 'quality' of the quantized signal is measured by the signal-to-quantization noise ratio (SQNR) which is mathematically written as:

$$SQNR = 10 \log_{10} \frac{P_x}{P_g} \qquad (3.5)$$

Where, $P_x$ and $P_q$ are the average powers of the DTCV and quantized signals respectively, and may be written as:

$$P_x = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2 \qquad (3.6)$$

And,

$$P_q = \frac{1}{N} \sum_{n=0}^{N-1} |x_q(n) - x(n)|^2 \qquad (3.7)$$

## EXERCISE:

**Task # 1:** Simulate a discrete time continuous valued (DTCV) sinusoid of 1/50 cycles/sample with length of the signal to be 500 & choose the no. of significant digits for round-off and apply to the signal generated above. Compute the error signals and SQNR.

**Task # 2:** For $f_o$ = 1/50 and N = 200, write a program to quantize the signal x (n), using

a) Truncation

b) Rounding to 64,128 and 256 levels. In each case plot the signals x (n), $x_q$ (n) and $e_q$ (n) and also compute the corresponding SQNR


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